

Enhancing Business Decision-Making Through Deep Learning Approaches for Time-Series Retail Demand Forecasting

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Abstract

The use of transactional data in large amounts has been a challenge in modern retail. This study uses the M5 transactional retail stores dataset to see whether the deep learning models that are advanced can help with business decision-making. The dataset is structured as a time series dataset which captures various transactional data from various retail store branches. There are three csv files that are used in this study that contain historical daily sales transaction data, calendar and transaction date information, and product's price information. The forecasting models that were used in this research includes 1D CNN, GRU, LSTM, DNN, and RNN. The forecasting models were evaluated by using WRMSSE, MAE, RMSE, and MAPE. The results show that the 1D-CNN model achieved forecasting performance that is the most consistent across WRMSSE, MAE, RMSE, and MAPE metrics. In contrast, recurrent architectures such as LSTM usually produced forecasting errors that are higher that is possibly due to its lower responsiveness to short-term demand fluctuations and abrupt changes in retail sales patterns. These findings suggest that local pattern extraction provided by convolutional operations is more effective for capturing the dynamic demand characteristics present in the M5 retail dataset.

Keywords

Time Series Forecasting, Deep Learning, Retail Demand Forecasting, M5 Dataset, Sequential Modeling

INTRODUCTION

Retail sales forecasting is complex due to many factors that are complex and dynamic. These factors are promotions, seasonality, holidays, changes in customer behavior, and changes in market trends. There are also seasonality patterns and long-term trends in the data that overlap. There is also time dependency in the retail sales data, where the past data affect the future data [1][2][3]. These factors make the retail sales data has patterns that are not stable.

The old statistical models are less capable in capturing non-linear patterns and long-term dependencies in the data. They struggle to learn and understand patterns from the retail sales data that is dynamic and often changes and ineffective in handling the complex interactions in the data such as pricing, promotion, seasonality, and changes in customer's preferences [4][5][6][7]. Different from the old statistical models, the deep learning models can learn and understand complex patterns that are based on time-series, can capture patterns that are from big data and historical data, and can understand the changes in demand patterns over time. They are also more accurate and more reliable in demand prediction, and more adaptive towards market changes, seasonal patterns that are complex, and promotion effects. Because of this, the deep learning models can be used for many products with demand patterns that vary [8][9][10].

Retail needs a forecasting model that can predict the stocks that have to be prepared and the product's price. The results of the model's predictions are used for business decision-making. If the predictions that are produced by the forecasting model are accurate then the decisions that are made are also accurate. If the forecasting model produces predictions that are not accurate, the retail store will face several problems. The problems include overstocking, stockout, increase in cost, the customer becoming disappointed, the customer's trust on the retail store decreases, financial loss [11][12][13]. The deep learning models can help retail stores to become more efficient. This is because the deep learning models can help the retail stores in managing the inventory, can give predictions that can help decide how many stocks that is optimal, decrease the risk of the retail stores having a stockout, and can give predictions that can help decide product's price accurately [14][15][16].

The proposed approach in this research is to compare the deep learning models that are used in this research in a fair way and in a way that does not only focus on the model's accuracy, but also analyze the comparative performance of the models across different product categories and regional retail patterns. The main contribution of this research is not only in comparing multiple deep learning architectures, but also in analysing the model behavior across various characteristics of retail demand that include demand that is sparse, sales behavior that is category-specific, and the variations of regional sales that are in the M5 forecasting dataset.

Previous studies on retail demand forecasting mainly focused on improving forecasting accuracy or evaluating a single deep learning architecture. Limited research has compared multiple deep learning models across different product categories, sales regions, and demand patterns in

large retail datasets such as M5. Therefore, this study compares several deep learning models based on both forecasting accuracy and performance across diverse retail demand conditions to support better business decision-making.

LITERATURE REVIEW

2.1. Traditional Time-Series Forecasting in Retail

Previous studies in retail use statistical time-series methods, such as ARIMA and exponential smoothing. The ARIMA can perform well in short-term prediction and can capture patterns that are stable, whereas the exponential smoothing is fast in performance and has a simple design. But, the ARIMA is less effective when the data often changes, whereas exponential smoothing struggles to capture complex patterns [17][18]. The old method works by assuming that the data has a linear pattern and is stable from time to time. Because of this, the old methods are dependent on simple data assumption. Because of this, the old methods are not fit to be used for modern retail data that often changes, is non-linear, and is affected by many factors [19][20].

2.2. Deep Learning Approaches for Time-Series Forecasting

Deep learning is effective for time series forecasting because the deep learning models can automatically learn patterns that are complex and non-linear. RNN, LSTM, and GRU models are designed to capture relationships between data based on time sequences. LSTM and GRU generally provide better prediction results than traditional RNNs on complex data. Previous research has shown that LSTMs have lower prediction error rates than traditional models such as ARIMA and standard RNNs. LSTMs are able to capture seasonality and long-term trends well. The drawback of LSTMs is that they require more computation and more careful parameter tuning. Recent research has developed bidirectional LSTMs and attention mechanisms to improve prediction accuracy. Hybrid models and attention mechanisms help the model focus on important information, resulting in more consistent and accurate predictions [21][22][23][24].

2.3. Deep Learning in Retail Demand Forecasting

Demand forecasting is used to estimate future product demand. The results of the forecasting help companies manage their product inventory. Forecasts that are accurate ensure that the product inventory is more aligned with the customer demand. Forecasts that are incorrect can

lead to overstocking or stockouts. Therefore, forecasting that is accurate is crucial to maintain efficient company operations [25][26][27].

The deep learning models are more accurate than the traditional statistical methods for forecasting retail demand. The deep learning models can capture patterns that are complex and non-linear in sales data. Deep learning models take into account factors such as promotions, seasonality, and changes in customer behavior. LSTM is used by many for time series data forecasting. LSTM can learn sequential relationships and long-term patterns from historical data. It produces more accurate demand predictions. It helps optimize inventory planning. It reduces the risk of overstocks and stockouts. It supports more effective business decision-making [28][29].

2.4. Previous Studies Using the M5 Forecasting Dataset

There is a study that develops forecasting models using the M5 forecasting dataset. This study compares the statistical models, the machine learning models, and the deep learning models. This study does not only focus on prediction accuracy, but also on the impact of the predictions to the inventory, such as cost and fill rate. The results of this study are that the LSTM model and the temporal CNN model have the highest performance overall. Based on the results, the LSTM model has the lowest RMSE which shows that the LSTM model produced predictions with high accuracy. The results also show that the temporal CNN has the lowest cost and highest fill rate. These models are superior because of these model's ability to capture temporal patterns that are complex and non-linear interactions in the data [30].

2.5. Research Gap

Previous studies on retail demand forecasting mainly focused on improving forecasting accuracy or evaluating a single deep learning architecture. Limited research has compared multiple deep learning models across different product categories, sales regions, and demand patterns in large retail datasets such as M5. Therefore, this study compares several deep learning models based on both forecasting accuracy and performance across diverse retail demand conditions to support better business decision-making.

METHODOLOGY/ DATA AND METHODS

3.1. Research Design

In the first stage, the M5 Forecasting dataset is prepared through several data preprocessing steps. First, the M5 Forecasting dataset is transformed from a wide format into a long format. After that, three tables that are from the M5 Forecasting dataset are combined into a single unified dataset. Then, the categorical variables that are in the dataset are converted into numerical variables. Finally, missing values that are in the dataset are handled.

In the second stage, the study develops several deep learning models. The deep learning models are LSTM, RNN, DNN, GRU, and CNN. Each deep learning model is tried out to see how well it can pick up patterns in the data.

In the final stage, MAE, RMSE, and MAPE are used. The results are compared with the actual data to see the difference. Based on this, the models can be compared to see which one gives better results.

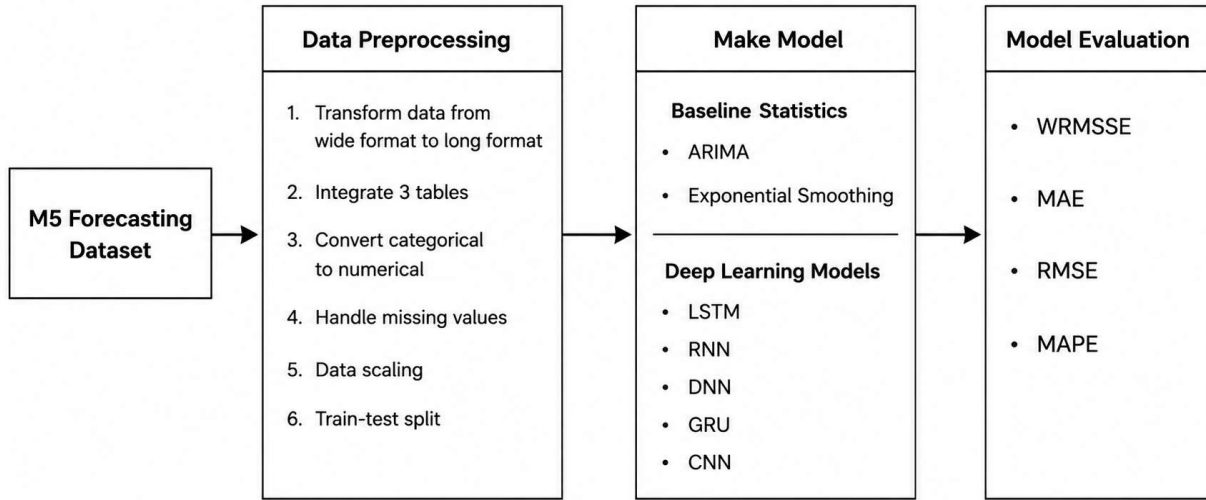


Fig. 1. Research Framework for Deep Learning-Based Retail Sales Forecasting

3.2. Dataset

The M5 forecasting dataset is taken from the M5 Accuracy Competition, and published through Kaggle as a best practice dataset for forecasting research [31]. The dataset is typically used to evaluate forecasting models.

To make a forecast that is more accurate, the dataset combines three primary data sources. These data sources are daily unit sales, weekly selling prices, and a comprehensive temporal

calendar. The features in the calendar of this dataset serve as critical and external factor variables. This is because the features in the calendar in this dataset include price fluctuations, seasonal trends, and special events. These events could have a big influence on consumer behavior. Then, the performance of the prediction model that has been made will be evaluated using the Weighted Root Mean Squared Scaled Error (WRMSSE) to account for the varying economic importance of different products across the hierarchical structure.

3.3. Data Preprocessing

3.3.1. Data Restructuring (Column-to-Row Transformation)

The dataset originally stored daily data in separate columns, which made it less practical to use. This kind of structure is in fact quite hard to use when dealing with time-based data. So, the data was rearranged into a long format using a melting process. After that, each row represents sales for one day.

3.3.2. Data Integration

The M5 dataset contains several CSV files. The CSV files are the sales_train_validation.csv, sales_train_evaluation.csv, sell_prices.csv, calendar.csv, and sample_submission.csv. In this research, the CSV files were combined into a single dataset so that all of the variables could be processed together during the forecasting stage. For joining each CSV, the merging was done using the wm_yr_wk column as a primary key to join sales_train_validation.csv, calendar.csv, and sell_prices.csv. This ensured that every sales record got the calendar date and the pricing details that are correct. After the data was combined, the models had access to things like calendar events and price changes, which are likely to affect customer demand.

3.3.3. Categorical to Numerical Transformation

The M5 dataset contains several variables that are categorical. The variables that are categorical are department categories, store locations, and state location. The variables that are categorical had to be transformed or converted. The variables that are categorical were converted into numerical variables to make the variables can be used to train the deep learning models that

can only process numerical data. In this research, label encoding was used to convert the variables that are categorical to numerical variables.

3.3.4. Data Cleaning and Missing Value Handling

After the process of data integration, the dataset that has been merged was examined for missing values to ensure completeness of the data before the model training. An analysis of missing value was performed on the resulting dataset (weekly_avg_enriched) after the data aggregation stage and the feature engineering stage. The results of the analysis showed that no missing values were present in any of the variables that are used for forecasting. Therefore, it is not required to do missing value imputation or record removal. The dataset that is complete was then used directly for model training and model evaluation.

3.3.5. Data Splitting

In the next step, the M5 forecasting that has been prepared and processed is divided into two sets, a training set and a testing set. This data splitting stage is done to evaluate the deep learning model performance that has not been seen before.

3.3.5. Preprocessing Results

Table 1. Dataset Characteristics Before and After Preprocessing

Characteristic	Before Preprocessing	After Preprocessing
Number of Records	30.490	1.233
Number of Features	1.919	6
Numerical Features	1.913	4
Categorical Features	6	2
Memory Usage	446.4 MB	57.9 KB
Data Format	Daily sales (wide format)	Weekly aggregated sales

The initial M5 dataset consists of 30,490 rows of data and 1,919 attributes (columns). The dataset contains product identity information and daily sales data from d_1 to d_1913. Most of the columns represent the sales amount for each day. The size of the dataset before data preprocessing

was approximately 446.4 MB. The structure of the data was still in wide format, with each day stored in a different column. This format is less efficient for the analysis process and for the process of the deep learning models training. That is why the data preprocessing stage was needed to be performed first.

After preprocessing, the data consisted of 1,233 rows with 6 columns. The columns that were used were `cat_id`, `state_id`, `wm_yr_wk`, `revenue`, `wday`, and `month`. The data that was originally in the form of daily sales, was converted to weekly sales data (weekly aggregation). Time features were added, such as day of the week (`wday`) and month (`month`). This was done to help the model recognize temporal patterns. The size of the dataset was reduced to approximately 57.9 KB. This size is significantly smaller than before preprocessing. The dataset that is the result of preprocessing is more compact, easier to be processed, and more suitable for training the deep learning models.

3.3.6. Data Transformation

The data scaling process is done by changing the data into a range from 0 to 1 by using `MinMaxScaler`. After that, the dataset is changed into a time-series dataset by using a function that is called `create_dataset`. Then, the data is reshaped so the data can fit the deep learning models that are used in this study.

3.4. Statistical Baseline Modeling

As a baseline for comparison in evaluating the performance of the deep learning models, this research used two traditional statistical forecasting methods, namely ARIMA (Autoregressive Integrated Moving Average) and Exponential Smoothing. These two methods were chosen because these two methods are usually used in time series forecasting and these two methods are often used as baselines in retail demand forecasting research. ARIMA is used to model temporal patterns and relationships between historical data, while Exponential Smoothing is used to capture level patterns, trends, and seasonality by giving weight that is greater to the most recent observations.

The ARIMA model and the Exponential Smoothing model were trained by using the same weekly data as the deep learning models so that the comparison can be done fairly. The results of the prediction from both baseline models were then evaluated by using the four evaluation metrics,

namely the WRMSSE evaluation metric, the MAE evaluation metric, the RMSE evaluation metric, and the MAPE evaluation metric. This comparison aims to know the extent to which the deep learning models can provide improved accuracy compared to the conventional statistical methods in forecasting product demand on the M5 Forecasting dataset.

This research focuses on comparing several deep learning architectures (1D-CNN, RNN, LSTM, GRU, and DNN) to analyze the ability of each deep learning model to complex retail demand patterns that are complex on the M5 Forecasting dataset. Although machine learning models such as XGBoost often show performance that is excellent, this research aims to evaluate deep learning approaches specifically so that the scope of the research remains focused and the results of the comparison between deep learning models can be analyzed in more depth.

3.5. Deep Learning Modeling

The deep learning models that are selected for this research are trained by using the dataset that has been prepared. The training process of the deep learning models is done separately for each state (CA, TX, and WI) to capture sales patterns and regional characteristics that are different. The deep learning models are trained by using the weekly revenue data for 50 epochs so that the models can learn temporal patterns more effectively and improve the performance of the forecasting. The deep learning models are trained by using a function that is called `plot_category`. This function takes data that is state-specific and takes weekly revenue information as input, then trains the deep learning models that are selected (DNN, 1D-CNN, RNN, LSTM, and GRU).

In this research, the deep learning models that are selected for this research (1D-CNN, GRU, LSTM, DNN, and RNN) are compared to determine which model performs best in forecasting retail demand. The `plot_category` function is executed three times, where each execution corresponds to one product category. The first execution is performed for the HOBBIES category, the second for the HOUSEHOLD category, and the third for the FOODS category. This approach ensures that each category has its own forecasting model and training process, allowing the models to better capture category-specific sales behavior and demand patterns.

To ensure a comparison that is fair, the hyperparameters of each deep learning model were optimized independently by using the Keras Tuner. During the process of hyperparameter tuning, the dataset was divided by using an 80:20 split for training and validation, where 80% of the data was used for training and 20% of the data was used for validation. The validation set was used to

evaluate various hyperparameter combinations and select the optimal configuration for each deep learning model. Different architectures, learning rates, layer configurations, and unit sizes were selected for each deep learning model based on the best tuning results. This approach allowed each deep learning model to be evaluated under its own configuration that is optimized rather than using hyperparameter settings that are identical across all architectures.

Table 2. Hyperparameters of Each Model

Hyperparameter	1D-CNN	GRU	LSTM	DNN	RNN
Learning Rate	$3.0e^{-3}$	$2.4e^{-3}$	$2.4e^{-3}$	$2.5e^{-3}$	$5.7e^{-4}$
Epochs	50	50	50	50	50
Batch Size	32	32	32	32	32
Kernel Size	3	-	-	-	-
Time Step	10	10	10	10	10
Number of Layers	1	2	1	3	1
Number of Units	96	96, 32	416	128, 32, 32	96
Loss Function	MSE	MSE	MSE	MSE	MSE
Activation Function	ReLU	Tanh, sigmoid, linear	Tanh, sigmoid, linear	Tanh, ReLU	Tanh, linear
Optimizer	Adam	Adam	Adam	Adam	Adam

Hyperparameter tuning was done by using Keras Tuner to identify the optimal configuration for each deep learning model. The best CNN architecture consisted of a single convolutional layer with 96 units and a learning rate of 0.0030. The GRU model that is optimal used two layers with 96 units for layer 0 and 32 units for layer 1 and a learning rate of 0.0024. The best LSTM configuration employed a single layer with 416 units and a learning rate of approximately 0.0024. For the DNN model, the optimal architecture included three layers with 128 units for layer 0, 32 units for layer 1, and 32 units for layer 2, using tanh and ReLU activation functions, and a learning rate of 0.0025. The RNN model achieved its best performance with a single layer that contains 96 units and a learning rate of 0.00057. All of the models were trained by using the Adam optimizer and Mean Squared Error (MSE) loss function.

3.6. Evaluation Approach

The forecasting performance in this research was evaluated by using multiple evaluation metrics. This is done to ensure that the accuracy of the evaluation of the deep learning models was comprehensive. The primary evaluation metric that was used in this research is the Weighted Root Mean Squared Scaled Error (WRMSSE), the official evaluation metric of the M5 Forecasting Accuracy competition, which takes into account the scale of each time series and the economic importance of products based on revenue contribution. WRMSSE was calculated by first calculating the Root Mean Squared Scaled Error (RMSSE) for each item-level series by using historical first-order differences as the scaling factor, then weighting each series based on total sales revenue that is obtained from the `sell_prices.csv` dataset, and aggregating the results across all item-store combinations over the forecasting horizon. A lower WRMSSE indicates better overall forecasting performance. Other than using the WRMSSE evaluation metric, the MAE evaluation metric, the RMSE evaluation metric, and the MAPE evaluation metric were also used in this research. The MAE evaluation metric was used to measure the average absolute prediction error, the RMSE evaluation metric was used to measure the squared prediction errors and is more sensitive to large errors, and the MAPE evaluation metric was used to measure the percentage of the prediction error. The use of several evaluation metrics helps give an evaluation of the model performance that is more comprehensive and more thorough.

RESULTS AND DISCUSSION

Table 3. ARIMA Model Evaluation Results of Each Product Type for Every State

State	WRMSSE			MAE			RMSE			MAPE		
	HO	HH	FO	HO	HH	FO	HO	HH	FO	HO	HH	FO
CA	2.94	2.70	1.61	2289.43	5468.65	10467.47	3197.52	7168.71	12442.10	13.38%	13.03%	14.27%
TX	3.19	4.33	1.51	2105.39	7039.82	7053.39	2515.07	8122.89	8505.33	19.75%	26.59%	13.49%
WI	1.67	3.10	1.67	1235.61	6458.72	11174.43	1430.31	6974.48	14226.37	15.59%	32.60%	23.82%

Table 4. Exponential Smoothing Model Evaluation Results of Each Product Type for Every State

State	WRMSSE			MAE			RMSE			MAPE		
	HO	HH	FO	HO	HH	FO	HO	HH	FO	HO	HH	FO

CA	1.02	1.06	0.88	943.72	3561.41	6274.65	1709.77	4542.99	8527.75	5.86%	7.60%	7.10%
TX	1.60	1.12	0.82	1504.36	2007.03	3991.71	1950.21	2995.08	5151.53	13.21%	7.68%	8.13%
WI	1.20	1.67	0.88	971.79	4473.89	7329.25	1373.64	5205.96	9105.36	12.59%	20.29%	14.68%

Table 5. WRMSSE Results of Each Product Type for Every State

State	1D CNN			GRU			LSTM			DNN			RNN		
	HO	HH	FO	HO	HH	FO	HO	HH	FO	HO	HH	FO	HO	HH	FO
CA	0.78	0.70	0.56	0.84	0.81	0.77	0.87	0.85	0.84	0.79	0.75	0.58	0.78	0.73	0.53
TX	0.74	0.73	0.46	0.83	0.80	0.54	0.87	0.82	0.80	0.77	0.72	0.53	0.78	0.78	0.45
WI	0.66	0.57	0.37	0.75	0.77	0.44	0.77	0.80	0.51	0.73	0.56	0.39	0.72	0.65	0.41

Table 6. MAE Results of Each Product Type for Every State

State	1D CNN			GRU			LSTM			DNN			RNN		
	HO	HH	FO	HO	HH	FO	HO	HH	FO	HO	HH	FO	HO	HH	FO
CA	561.54	1094.66	3029.68	617.34	3937.11	8744.99	631.53	1312.52	4732.93	584.65	1205.12	3207.77	573.53	1142.93	2930.68
TX	364.93	871.60	1829.84	421.58	2664.30	6621.57	454.20	1019.27	3576.20	386.65	850.89	2161.16	399.92	973.15	1738.61
WI	382.56	860.60	2335.57	444.54	2611.20	8201.26	457.80	1311.00	3435.77	437.20	870.66	2569.29	445.75	1055.90	2693.89

Table 7. RMSE Results of Each Product Type for Every State

State	1D CNN			GRU			LSTM			DNN			RNN		
	HO	HH	FO	HO	HH	FO	HO	HH	FO	HO	HH	FO	HO	HH	FO
CA	741.42	1411.35	3986.41	796.61	5055.85	11869.30	827.59	1701.53	6046.11	748.31	1508.26	4162.12	739.34	1466.05	3769.35
TX	487.35	1129.39	2489.43	543.23	2928.99	7363.39	569.31	1278.65	4315.75	506.85	1117.23	2838.39	509.79	1215.43	2392.26
WI	523.31	1166.46	3118.59	599.42	3045.36	8785.25	611.83	1635.96	4375.42	580.46	1148.17	3333.68	572.06	1331.50	3522.94

Table 8. MAPE Results of Each Product Type for Every State

State	1D CNN			GRU			LSTM			DNN			RNN		
	HO	HH	FO	HO	HH	FO	HO	HH	FO	HO	HH	FO	HO	HH	FO
CA	3.75%	2.87%	3.83%	4.10%	3.37%	5.53%	4.22%	3.45%	6.05%	3.91%	3.18%	4.11%	3.84%	2.99%	3.74%
TX	3.71%	3.57%	3.74%	4.25%	4.04%	4.64%	4.57%	4.15%	7.28%	3.94%	3.52%	4.42%	4.05%	3.99%	3.53%
WI	4.70%	3.97%	5.11%	5.48%	5.66%	6.57%	5.65%	5.90%	7.73%	5.34%	4.01%	5.56%	5.55%	4.88%	5.89%

Notes: HO = Hobbies, HH = Households, FO = Foods

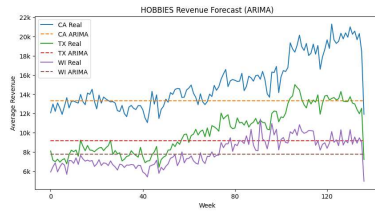
Based on Table 3 and Table 4, the ARIMA model and the Exponential Smoothing model showed forecasting performance that is relatively lower compared to the deep learning models in almost all product categories and regions. Although the Exponential Smoothing model generally produced error values that are lower than the ARIMA model, both statistical models consistently recorded higher WRMSSE, MAE, RMSE, and MAPE values than the deep learning approaches. This shows that the traditional forecasting methods have limited capability in capturing the complex, dynamic, and non-linear demand patterns present in the M5 retail dataset. While ARIMA relies on linear temporal relationships and Exponential Smoothing emphasizes recent observations, neither model is able to effectively model the interactions between seasonality, promotions, and changing consumer behavior. As a result, both the ARIMA and the Exponential Smoothing models are underperformed compared to the deep learning models, which are better suited for learning patterns that are complex from retail time-series data that is large-scale.

As shown in Tables 5–8, the ID-CNN model usually has forecasting errors that are lower, such as WRMSSE, MAE, RMSE, and MAPE, and the 1D-CNN model works well in most of the product categories and states, especially in the Hobbies (HO) category. Sometimes, the RNN model has error values that are slightly lower, but the difference is small, for example a lower WRMSSE of 0.53 in California (CA) and 0.45 in Texas (TX) for the Foods category. This means that the sales data in the M5 forecasting dataset has small patterns that are repeated, and the ID-CNN model can learn these patterns well by using the convolutional operations, so the 1D-CNN model can give forecasting results that are more stable in different retail categories.

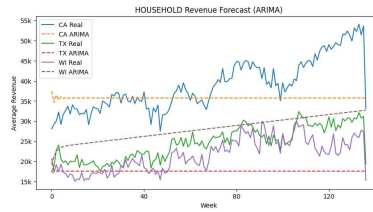
According to Table 6 and Table 7, the results of MAE and the results of the RMSE show that the ID-CNN model usually gives predictions that are closer to the real sales values compared to the GRU model and the LSTM model. On the other hand, the GRU model and the LSTM model often have forecasting errors that are larger in many categories and states. One possible reason is that the retail demand in the M5 forecasting dataset has continuous short-term changes and different sales patterns, so memory-based models find it hard to learn similar patterns across categories. This can be seen clearly in the large error values in unstable categories. For example, the MAE for Foods (FO) in California is 8744.99 for GRU and 4732.95 for LSTM, while ID-CNN is much lower at 3029.68. Because of this, GRU and LSTM may learn sequential relationships with different levels of effectiveness in different categories, which leads to less consistent results. Therefore, models that focus on capturing main local patterns using convolutional layers may work better in this forecasting task.

Tables 6–8 show that the results of the prediction change depending on the product category and the state. For example, in the Foods (FO) category in Texas (TX), ID-CNN gets a low MAPE of 3.74% compared to GRU (4.64%) and LSTM (7.28%), but the RNN model is even lower at 3.53% in this case. In other areas, the differences between the models are very small. This shows that different product groups have different shopping patterns, so some are easier to predict than others. Even so, the ID-CNN deep learning model still has error values that are low in almost all cases. This shows that the performance of the 1D-CNN deep learning model is more stable and more consistent than the other deep learning models.

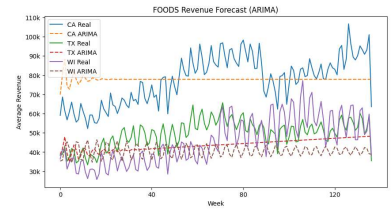
The results in Tables 5–8 also show that no deep learning model is always the best in every category and every measurement. This means that the prediction of the retail sales depends on each product's pattern. Some deep learning models perform better only in certain cases, but the ID-CNN deep learning model is more consistent overall, especially in the Hobbies (HO) category. This shows that for retail data that often changes like the M5 forecasting dataset, focusing on local patterns that are simple can give results that are more reliable and more stable.



(a)

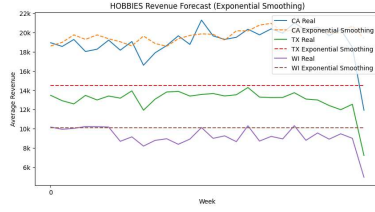


(b)

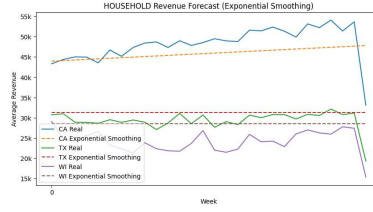


(c)

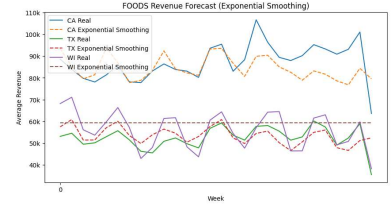
Fig. 2. ARIMA Predictions Comparison: (a) Hobbies; (b) Household; (c) Foods



(a)

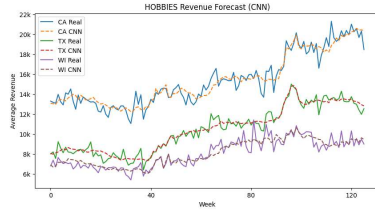


(b)

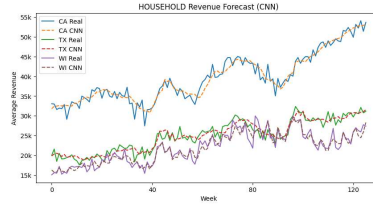


(c)

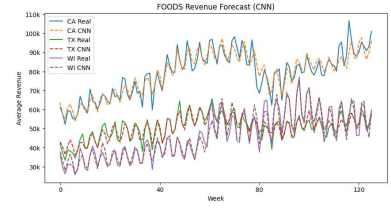
Fig. 3. Exponential Smoothing Predictions Comparison: (a) Hobbies; (b) Household; (c) Foods



(a)

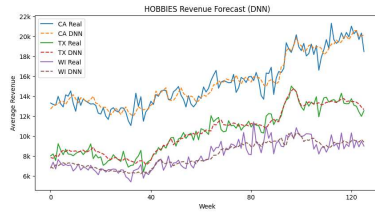


(b)

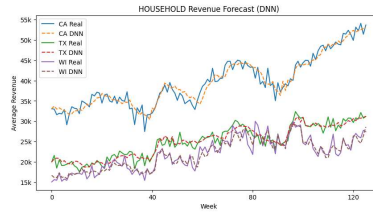


(c)

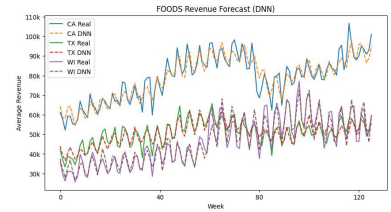
Fig. 4. CNN Predictions Comparison: (a) Hobbies; (b) Household; (c) Foods



(a)

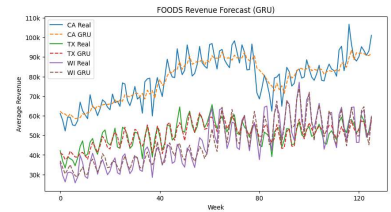
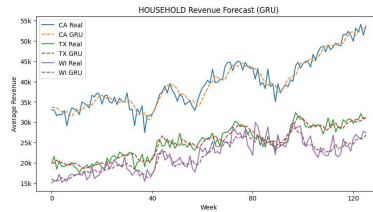
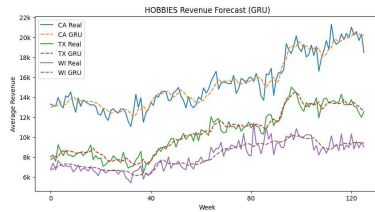


(b)



(c)

Fig. 5. DNN Predictions Comparison: (a) Hobbies; (b) Household; (c) Foods



(a)

(b)

(c)

Fig. 6. GRU Predictions Comparison: (a) Hobbies; (b) Household; (c) Foods

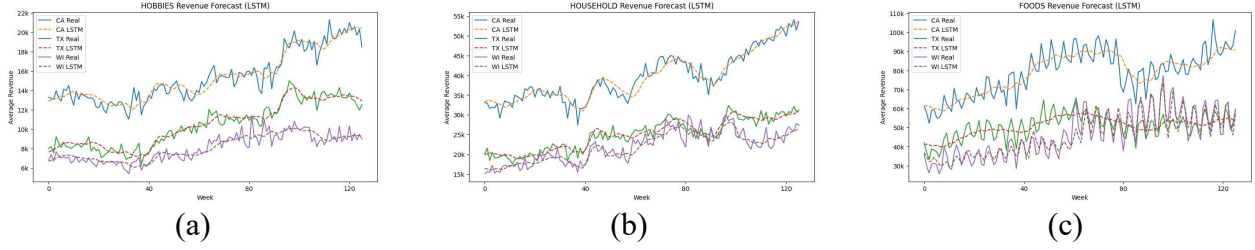


Fig. 7. LSTM Predictions Comparison: (a) Hobbies; (b) Household; (c) Foods

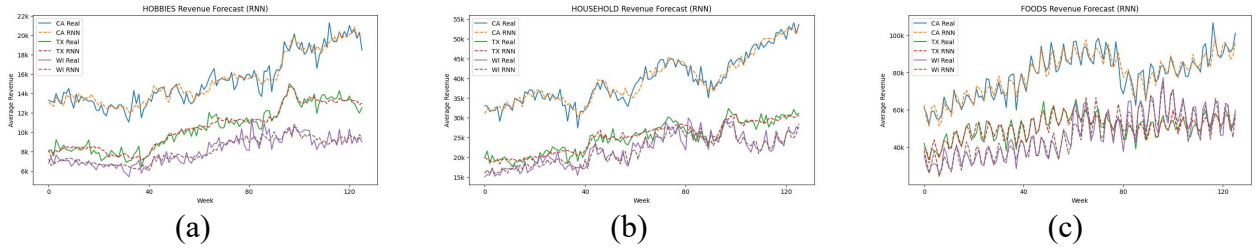


Fig. 8. RNN Predictions Comparison: (a) Hobbies; (b) Household; (c) Foods

The figures above show comparisons between actual sales data and the predictions across forecasting models—namely CNN, DNN, GRU, LSTM and RNN. The models cover various categories of items, which are Hobbies, Household and Foods through three states: California, Texas and Wisconsin.

The 1D-CNN model shows strong localized pattern extraction because of its architecture where it allows hierarchical data processing structure to occur. It uses segmentation where data input has a fixed temporal window with a fixed size of ($x_{\text{window}} = 10$) of continuous historical weeks, allowing it to learn from localized baselines instead of historical trends. In the next layer, convolutional feature extraction, the model passes 64 filters through a kernel width of 2 across the 10 week matrix using a rectified Linear Unit (ReLU) activation. Finally, in the max pooling downsampling layer with a set stride and pool size of 2 is applied directly after the convolution step. This extracts only the peak structural features from the local feature maps and removes variances that are not of significance.

The results from (Fig. 4 (a), (b), (c)) captures 1D-CNN characteristics appropriately through visuals of the prediction outcome based on MAE, RMSE, MAPE results. Throughout the observation of (Fig. 4) the prediction line is shown time and time again that it doesn't contain the

same significant formation of teeth-like fluctuations, unlike the lines from the actual data. This is the byproduct of spatial downsampling via the Max Pooling layer, the layer that filters out high-frequency stochastic noise, thereby stabilizing standard error metrics such as Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) and Mean Absolute Percentage Error (MAPE), explaining the occasional smooth line in some parts of the data that seems to have supposedly many inconsistent spikes, yet still follow the general incline and decline of the line. With all of these factors playing together in producing such results, amongst other models in error occurrence, structure of the shape following real data patterns and appropriate amount of details of fluctuations makes 1D-CNN to be rated to have the most stable prediction results.

In contrast, LSTM focuses on long-term patterns that have occurred over the past few periods. Because of this LSTM predictions appear smoother and follow the general trend direction. However, the LSTM model is slower in responding to changes in sales that are sudden. This causes predictions lag and prediction errors that are larger, especially in retail data where sales patterns often fluctuate. This can be seen from the results of WRMSSE, MAE, RMSE, and MAPE evaluations, where the error values of the LSTM model are usually higher than those of the 1D-CNN model. Therefore, on the M5 dataset, LSTM is less effective at capturing short-term demand fluctuations, while 1D-CNN is better able to track dynamic changes in sales patterns.

CONCLUSION AND FUTURE RESEARCH/ CONCLUSION AND POLICY RECOMMENDATIONS

This research studied several deep learning models through the M5 Walmart dataset to address the forecasting problems that are usually unstable on a daily basis. Based on the research test result, 1D-CNN has the highest accuracy level, which means that 1D-CNN can handle issues such as unpredictable demand and can also be used to capture data patterns well over various periods, so 1D-CNN can be used for future research as the baseline, because the model can produce more stable and accurate results than the others models. For future consideration, a model that is made by combining two models, CNN and LSTM, can be used to improve the prediction results for the M5 competition dataset. These results can be used to help select reliable models for better demand forecasting.

AI USAGE DECLARATION

AI tools were used to support writing and language refinement. The authors conducted and verified all research, analyses, and conclusions presented in this study.

AUTHOR CONTRIBUTION & OPEN DATA

Author A coordinated the research project, wrote the introduction, abstract, and discussion, and acted as Project Coordinator. Author B developed the research methodology, developed and implemented the model, and wrote the discussion and conclusion. Author C evaluated and analyzed model performance, contributed to the interpretation of model evaluation results, and contributed in the development of visualizations and analysis of research results.

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Author Biographies

Author A is a Computer Science undergraduate student at a university. He has interest in machine learning, deep learning, data analytics, and in applying computational methods to address and solve problems in the real world. His academic experience consists of working on projects that are based on data and working on data engineering projects that involve data preprocessing, feature engineering, and scalable data processing pipelines. He is also familiar with clustering technique and classification technique, and has direct experience in using tools such as Python, PySpark, and data science libraries that are popular. Outside of the academic activities at his university, he is involved in exploring and learning about how artificial intelligence can be applied across various fields, such as business intelligence, e-commerce analytics, and predictive modeling.

Author B is an undergraduate student that majors in Computer Science at a university. His academic interests focus on data science, machine learning, artificial intelligence, and the application of computational methods to address and solve the problems in the real world. He chose the Database Technology stream because of his interest in working with data, his interest in identifying patterns that are meaningful, and his interest in utilizing analysis that is based on data to gain insights that are valuable and useful for solving problems and making business decisions in the real world. Through various academic projects, he continues to enhance both his knowledge and practical skills in data-related fields. Through academic activities and activities that are project-based, he has developed experience in data preprocessing, experience in predictive analytics, experience in developing deep learning models, and experience in database systems.

Author C is an undergraduate student that majors in Computer Science at a university. She has a good academic background in computer science with a focus on database technology and computational problem-solving. Her primary research interests are in computer science, especially in areas that are related to data analysis, machine learning, and the application of computational methods to problems in the real-world. She has participated in academic research activities and has published a paper in a computer science competition under LPB (Lomba Peneliti Belia). The paper is titled “Utilization of the ‘Children’s Nutrition Standards’ Application in order to Monitor Developments and Nutritional Growth of Children Aged 5 to 18 Years based on Body Mass Index.” Through this work, she gained experience in developing computer applications for healthcare and real-world problem-solving.